



Spar disbond visualization in in-service composite UAV with ultrasonic propagation imager



Jung-Ryul Lee^{a,b,*}, Chang Min Cho^c, Chan Yik Park^c, Chung Thanh Truong^a,
Hye Jin Shin^d, Hyomi Jeong^d, Eric B Flynn^e

^a Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology, Republic of Korea

^b X-NDT Inc., Republic of Korea

^c Aeronautical Technology Directorate, Agency for Defense Development, Republic of Korea

^d LANL-CBNU Engineering Institute-Korea, Chonbuk National University, Republic of Korea

^e The Engineering Institute, Los Alamos National Laboratory, United States

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ABSTRACT

Composite materials have been increasingly used for aircraft structures due to their major advantage of being lightweight compared to metallic materials. However, the drawback of composite materials is that they easily sustain disbond damages due to load and impacts during manufacture or service. An effective quality control management system for aircrafts is required for early detection and early response to such critical damages. This paper reports the application of the Ultrasonic Propagation Imager (UPI) for damage inspection of an in-service aircraft. The inspection task took place at a Korean air force base in May 2013 with the objective of determining the structural condition of the composite aircraft wing at various areas where disbond damages were suspected. The existence of many structural features such as multiple rivets and spars complicated the task since those additional structural features interfere with the laser ultrasonic waves. By developing a novel wavenumber domain filtering algorithm, we successfully detected the disbond damages on the aircraft wing. This result proved the feasibility of the UPI to serve as an effective structural health management system for real-world aircraft applications.

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1. Introduction

During the last several decades, there has been a substantial shift in aircraft structures from traditional materials such as steel or aluminum to composite materials [1]. The advantages of composite materials are numerous, such as low density and high stiffness, which result in weight savings; the ability to employ microstructure design for different purposes; and also since multiple parts of the aircraft structures can be combined in a single cure, there is a significant reduction in the need for connections such as rivets or fasteners, which again results in weight savings. However, one major downside of composite materials is that they are susceptible to impact damages and disbond [2]. Conventional non-destructive testing methods, such as ultrasonic, X-ray, thermography and eddy current methods require heavy, complicated

equipment, and are labor-extensive and time-consuming, especially for such large-scale structures. The aeronautic industry therefore is still intensively searching for an effective quality control management system for early detection and early response to such critical damages.

This paper reports the inspection results of using the Ultrasonic Propagation Imager (UPI) [3] on an in-service composite aircraft wing, with the focus of introducing a novel wavenumber domain filtering algorithm for damage detection. This class of signal processing algorithm has been an active research topic in recent years with the time history wavefield obtained using scan laser Doppler vibrometers [4], air coupled transducers mounted on a scanning stage [5], or Q-switched laser scan systems [3]. In the pioneer paper [6] on this topic, Ruzzene processed the wavefield through directional filtering in the frequency wavenumber domain, which allows for removal of incident waves and highlighting of the reflection waves associated with damages. Michaels et al. [7] applied a three-dimensional Fourier transform for mode separation and analysis of reflection and mode conversion due to notch defects. Kudela et al. [8] showed that energy wave mapping after processing with wavenumber domain filtering is effective for visualizing damages.

* Corresponding author at: Department of Aerospace Engineering, Korea Advanced Institute of Science and Technology, 291 Daehak-ro, Yuseong-gu, Daejeon, 305-701, Republic of Korea.

E-mail address: leejrr@kaist.ac.kr (J.-R. Lee).

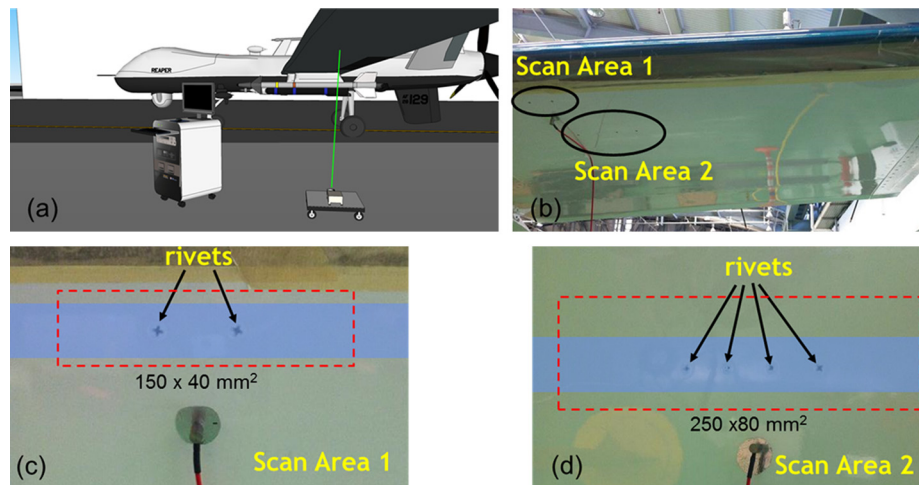


Fig. 1. (a) Aircraft composite wing inspected using the UPI system, (b) two inspection areas on the wing, (c) detail of scan area 1, (d) detail of scan area 2.

Flynn et al. [9] mapped the wavenumber by isolating a single wave mode and estimating the wavenumber at a particular frequency that maximized the wave energy at each grid scanning point. Yu and Tian [10] used a short-space two-dimensional Fourier transform to obtain the space–frequency–wave number representation and to study the wave interaction with structural discontinuity and crack detection on an aluminum plate.

The proposed wavenumber filtering algorithm is as follows. First, the adjacent waves subtraction algorithm [11] was applied to the measurement wavefield. This process suppressed incident waves and exaggerated the anomalous waves caused by damages. Next, since the wavenumber of the ultrasonic waves changes when the waves propagate through structural inhomogeneity such as geometry discontinuities or disbond, by applying a narrowband wavenumber filter, further isolation of the waves associated with the damages was achieved. As a result, this algorithm eliminates strong incident waves that obscure the weak scattering waves due to damages. Thus when applying the wave energy mapping in the next step, it is able to show the damages together with structural features of the inspected areas.

This paper is organized as follows. In the next section, the test structure: an in-service aircraft composite wing, and the non-destructive evaluation (NDE) methodology of the UPI system are introduced. Next, the proposed wavenumber filtering algorithm is presented and applied to the test structure, which was successfully detected the disbond damages. Finally, the effectiveness of the proposed algorithm is confirmed with an inspection test of a specimen with known artificial damage.

2. Test structure and the NDE methodology of the UPI system

The inspection task took place at a Korean air force base in May 2013 with the objective of determining the structural condition of the composite aircraft wing at various areas where disbond damages were suspected. The test structure is shown in Fig. 1(a) and (b). Scan area 1 was 150 mm width and 40 mm height while scan area 2 was 250 mm width and 80 mm height, both with the same scan interval, i.e., the distance between two adjacent scan points of 0.25 mm. The disbond damages between the wing skin and spar was attempted to fix using rivets as shown in Fig. 1(c) and (d): two rivets in scan area 1 and four in scan area 2. The color band areas, as seen in Fig. 1(c) and (d), were intentionally drawn to emphasize the location and size of the spar area. In the actual structure, the 30 mm-height spar lies behind the wing skin and therefore cannot be seen from outside.

The Ultrasonic Propagation Imager (UPI) [3] is a viable candidate to serve as an effective structural health management system for real-world aircraft applications. The core of the UPI is a high speed Q-switched laser and a mirror scanner as shown in Fig. 2. The pulsed laser beam is rapidly excited at each point of a predefined scan grid and the laser ultrasonic signal is measured at one or many fixed sensing point(s).

The UPI has several advantages. First, it has very high-speed scan. The latest generation of the UPI system has reached a scan speed of 20,000 points per second [3], which means that the inspection time for an area of 1 m × 1 m with spatial resolution of 1 mm is less than 1 min. Second, the UPI is not fixed to any particular sensing system but can be hooked up to a wide range of sensor types, from contact sensors such as piezoelectric transducers and optical fiber sensors to non-contact sensors such as laser Doppler vibrometers and air-coupled transducers. This characteristic distinguishes the UPI system from many other non-destructive testing systems where applications are largely limited by the physical form and sensitivity of the sensing part. Third, the UPI is portable, as it is able to operate at a far distance from the inspection target, up to 100 m using a beam collimator. Fourth, the inspection results of the UPI are completely reference-free, in the sense that there's no need for "baseline data" of the non-damaged state of the structure. Since the virtual wavefield is represented as a movie, it is intuitive for damage realization. Finally, the UPI is easily extended by using a set of electric-controlled mirrors to alter the direction of the laser beam for inspecting structures with very large areas.

The UPI system with capable of maximum 1 kHz pulse repetition rate was used for this test. This system has a scan speed of 1000 data points per second. The laser was a 532 nm Q-switch laser with approximately 1.3 mJ pulse energy. Integrated piezoelectric transducers were used to measure the laser ultrasonic waves. The sensors were installed at the center line of the scan width and 20 mm below the lower limit of the scan area. The laser ultrasonic waves were passed through a band-passed filter with a range of 10 kHz–500 kHz and amplified with a gain at 30 times the original signal. Finally, the signals were collected using a digitizer installed in a CPU.

3. Wavenumber domain signal processing

3.1. Full wavefield measurements

The full scanning data forms a three-dimensional matrix $V \times H \times N$, where V and H represent the number of scanning points

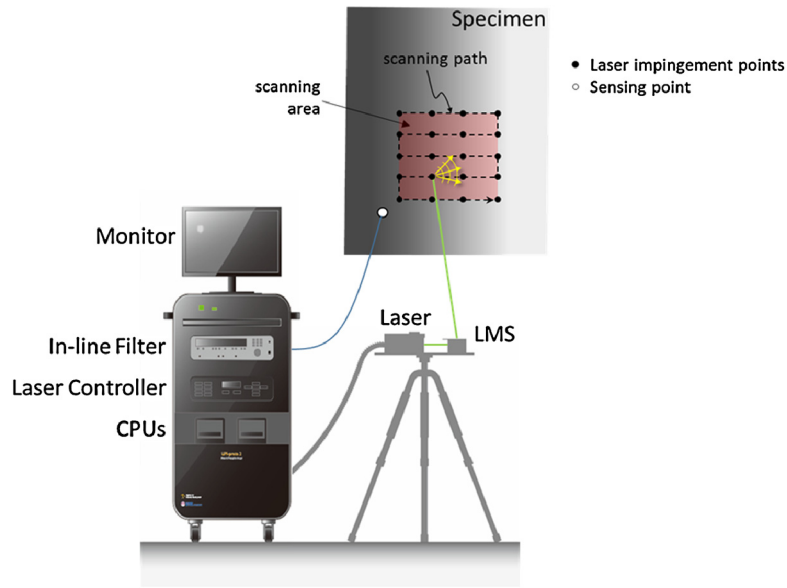


Fig. 2. Experimental setup with Ultrasonic Propagation Imager (UPI) (LMS: laser mirror scanner, CPU: central processing unit).

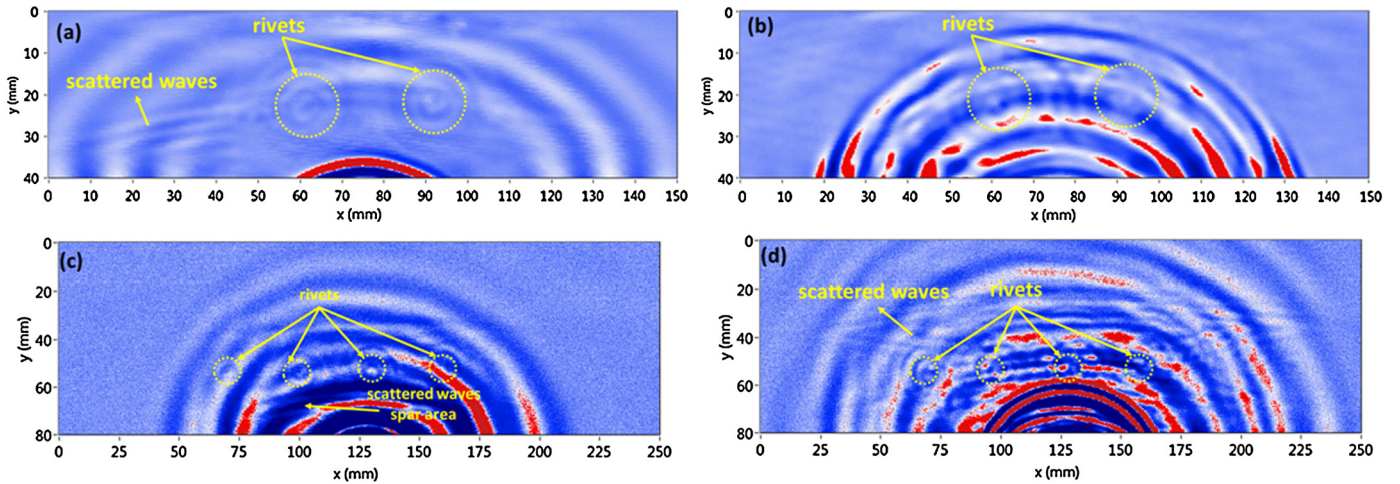


Fig. 3. Ultrasonic waves propagation image of (a) scan area 1 at 20.8 μ s, (b) scan area 1 at 44.0 μ s, (c) scan area 2 at 20.8 μ s, and (d) scan area 2 at 30.4 μ s.

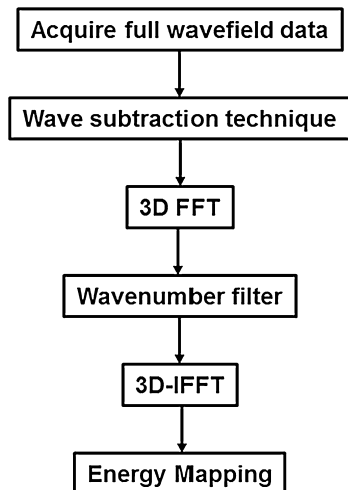


Fig. 4. Flowchart of the proposed algorithm for the visualization of composite spar disbond.

in the vertical and horizontal directions of the scanning area, while N is the sampling length of the measurement data for each scanning point. A slide perpendicular to the N axis maps the ultrasonic wave amplitude of the scan area at a particular time point. Plotting successive slides along the time axis generates a movie of ultrasonic waves propagation.

Fig. 3(a) and (b) show the ultrasonic waves propagation slides of scan area 1 at 20.8 μ s and 44.0 μ s. The scattering waves from the rivets as well as those from the spar area are easily observed. However, this basic waves propagation mapping does not provide evidence for damage assessment of this structure. The same conclusions are applied to scan area 2 where Fig. 3(c) and (d) show ultrasonic waves propagation slides at 20.8 μ s and 30.4 μ s. We will show that by digging deeper into the rich wavefield with an advanced signal processing method using wavenumber domain filtering, we are able to detect the disbond damages.

3.2. Proposed wavenumber filtering algorithm

The flowchart of the proposed algorithm is shown in Fig. 4. After acquiring the full wavefield data, the first step is to apply the

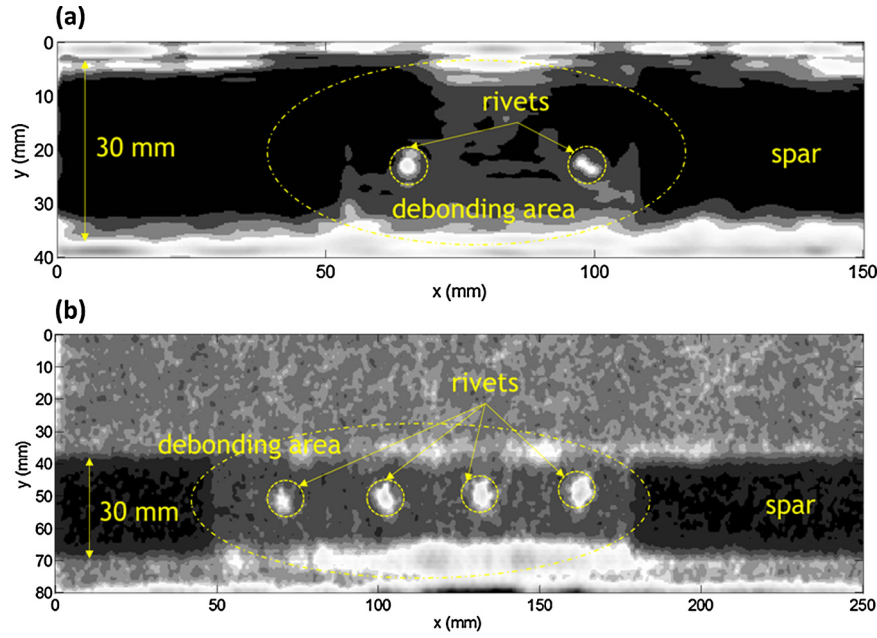


Fig. 5. Wavefield energy maps using the proposed algorithm for (a) scan area 1, and (b) scan area 2.

adjacent waves subtraction algorithm [11] to the wavefield measurement. In previous papers [12], this algorithm has proven to be very effective in detecting several type of damages (stringer tip disbond, skin-spar disbond, and invisible impact damage) on a composite wing. The waves of two spatial adjacent points are matched in phase and then subtracted from each other. Mathematically, this process is presented using equations (1) and (2) as follows

$$r_{s_i, s_{i-1}}(l) = \sum_{n=l}^{N-l-1} s(n)_i s(n-l)_{i-1} \quad (1)$$

$$\Delta s(n_i) = s(n)_i - s(n + l_{r_{\max}})_{i-1} \quad (2)$$

where r denotes the cross correlation between two adjacent signals $s(n)$ at points i and $i-1$; n is the index of the discrete-time instants, N is the number of sampling points for each signal, and l is the time shift. The time shift that maximize the cross correlation r , denoted by $l_{r_{\max}}$, is subsequently used for calculating the adjacent waves subtraction signal $\Delta s(n_i)$ as shown in equation (2). This process suppresses the incident waves and exaggerates the scattering waves caused by the damages [11].

Through the waves subtraction process, as shown in equations (1) and (2), the original wavefield $u(x, y, t)$ is transformed to the adjacent waves subtracted wavefield $\tilde{u}(x, y, t)$. In the next step, a wavenumber domain filtering algorithm is applied. The wavefield $\tilde{u}(x, y, t)$ is transformed from the space-time domain to the wavenumber-time domain $\tilde{U}(k_x, k_y, t)$ using the two-dimensional Fourier transform as shown in equation (3)

$$\tilde{U}(k_x, k_y, t) = F_{2D}[\tilde{u}(x, y, t)] \quad (3)$$

where k_x and k_y are the x -direction and y -direction wavenumber, respectively.

Next, a wavenumber Gaussian filter [8,9] is applied to $\tilde{U}(k_x, k_y, t)$. The result is a narrow-band wavefield in the wavenumber domain $U_F(k_x, k_y, t)$

$$\tilde{U}_F(k_x, k_y, t) = \tilde{U}(k_x, k_y, t) \Phi(k_0, \sigma) \quad (4)$$

where $\Phi(k_0, \sigma)$ is a Gaussian wavenumber mask at a wavenumber central k_0 and standard deviation σ expressed as

$$\Phi(k_0, \sigma) = \exp\left(\frac{-(\sqrt{k_x^2 + k_y^2} - k_0)^2}{2\sigma^2}\right) \quad (5)$$

The inverse Fourier transform is then applied to transform $\tilde{U}_F(k_x, k_y, t)$ back to the space time domain as shown in equation (6). Finally, the wave energy [8] at each grid point of the scan area is calculated using equation (7) as follows

$$\tilde{u}_F(x, y, t) = F_{2D}^{-1}[\tilde{U}_F(k_x, k_y, t)] \quad (6)$$

$$\tilde{E}(x, y) = \int_0^T \frac{1}{2} [\tilde{u}_F(x, y, t)]^2 dt \quad (7)$$

3.3. Damage detection results

Fig. 5(a) shows the wavefield energy maps using the proposed algorithm for scan area 1. The spar area strongly bonded to the wing skin showed a high energy level of ultrasonic waves, unlike the section that debonded from the wing skin. The same observation applies to Fig. 5(b), corresponding to scan area 2. The results show that the proposed algorithm successfully detected the disbond damage in the inspected structure.

The selection of the central wavenumber k_0 is critical for proper isolation of the disbond damage features. Different values of the central wavenumber correspond to different layers of the wavefield. The wavenumber-frequency curve could provide some hints for selection of the central wavenumber k_0 . Fig. 6(a) shows the wavenumber frequency curve of scan area 1 and Fig. 6(b) shows the wavenumber frequency curve of scan area 2. The central wavenumber was selected where there were significant “spurious” points deviated from the main curve S_0 or A_0 mode. This was done because wave propagation through a damaged area follows a different wavenumber-frequency characteristic. For this particular study, the central wavenumbers applied to scan areas 1 and 2 were 0.09 mm^{-1} and 0.14 mm^{-1} , respectively. Further study is on-going to develop an approach to automatically select this parameter for certain types of damages in the structure.

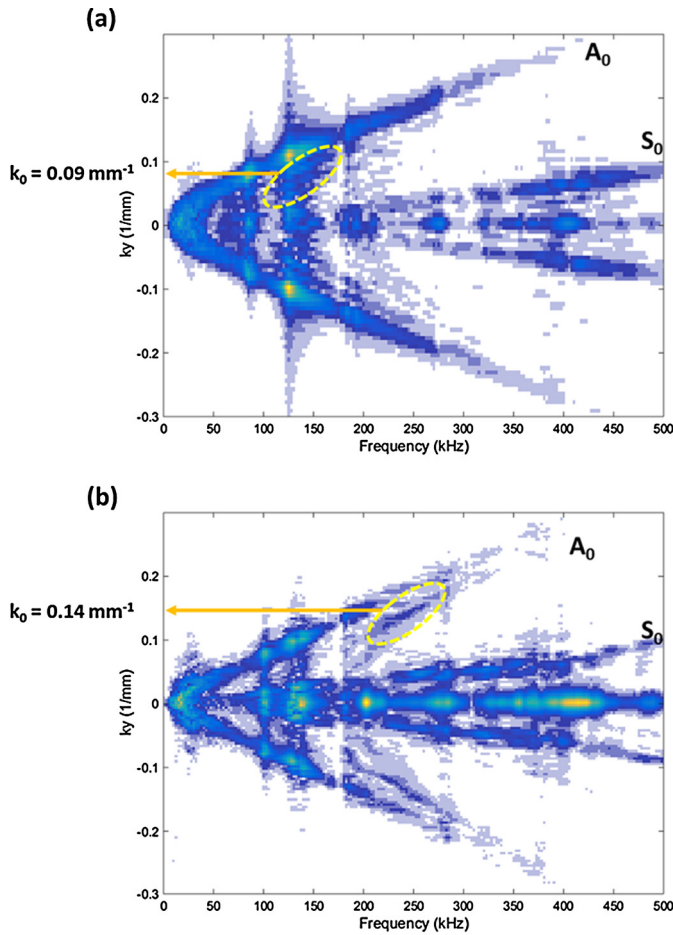


Fig. 6. Wavenumber frequency magnitude map of (a) scan area 1, and (b) scan area 2.

4. Validation test of the proposed algorithm with an artificial disbond damage specimen

Due to limited access to the military in-service aircraft, there was no available C-scan result from a previous test to confirm the validity of the proposed algorithm. Therefore, we set up a validation test with known disbond damage. Fig. 7(a) and (b) shows that the test specimen was made by two identical carbon fiber-reinforced plastic (CFRP) plates size 400 mm × 800 mm, each with thickness of 1.5 mm, attached together using adhesive tape

with thickness of about 0.5 mm. Artificial disbond damage was generated by cutting the adhesive layer with an area of 20 × 20 mm². The scan area was 100 × 100 mm² and the laser ultrasonic signal was detected using an integrated piezoelectric transducer.

Fig. 8(a) and (b) show the waves propagation images at 63.2 μs and 72.8 μs, respectively. The boundary of the disbond damage is apparent where the front waves changes curvature. The proposed algorithm was applied to the wavefield to detect the disbond damage and the result is shown in Fig. 9(a). The selected central wavenumber was 0.1 mm⁻¹. The artificial disbond area trapped the energy and showed a strong waves energy field with the width of 18 mm and height of 21 mm. The C-scan result is shown in Fig. 9(b) where the damage size is as expected at 20 × 20 mm². This result confirmed the effectiveness of the proposed wavenumber domain filtering algorithm.

5. Conclusions

This paper studied the applicability of the Ultrasonic Propagation Imager (UPI) to detect disbond damages on an in-service composite wing aircraft. The laser ultrasonic wavefield obtained using the UPI was analyzed with a modified wavenumber domain filtering technique and was successful in detecting the disbond damages. The proposed algorithm was then confirmed by C-scan results with a validation test on a specimen containing known artificial disbond damage. These results have proved the feasibility of the UPI system to serve as an effective structural health management system for real world aircraft applications.

Conflict of interest statement

These authors declare that this article does not have any conflict of interest.

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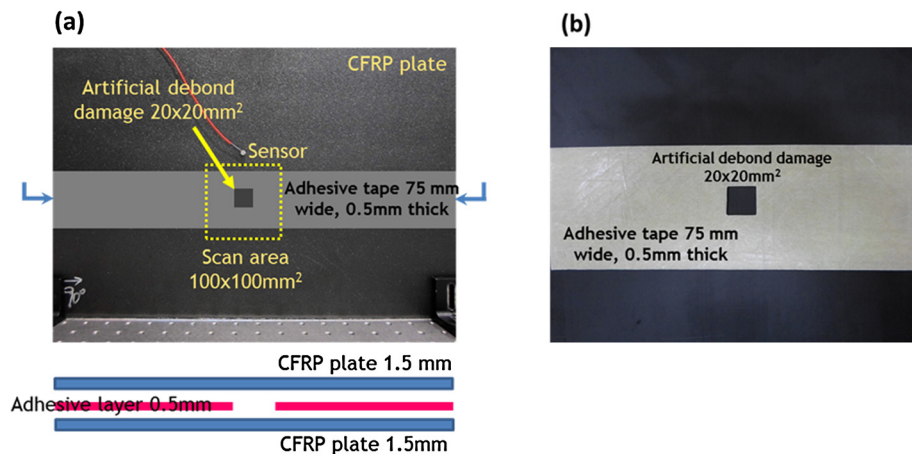


Fig. 7. Validation test for the proposed algorithm: (a) detail of the scan area and the cut section, (b) detail of the adhesive tape and the artificial cut with size of 20 × 20 mm² before attachment to the second CFRP plate.

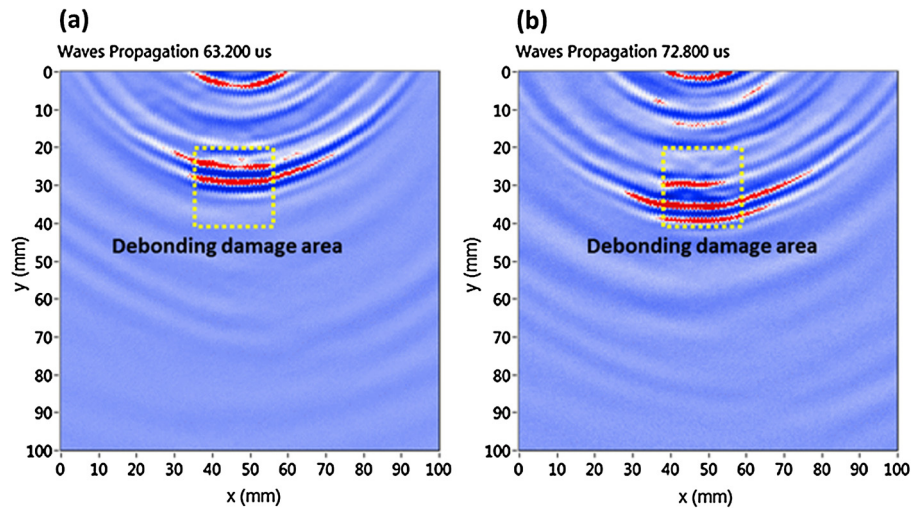


Fig. 8. Ultrasonic waves propagation image of the specimen with artificial disbond damage at (a) 63.2 μ s and (b) 72.8 μ s.

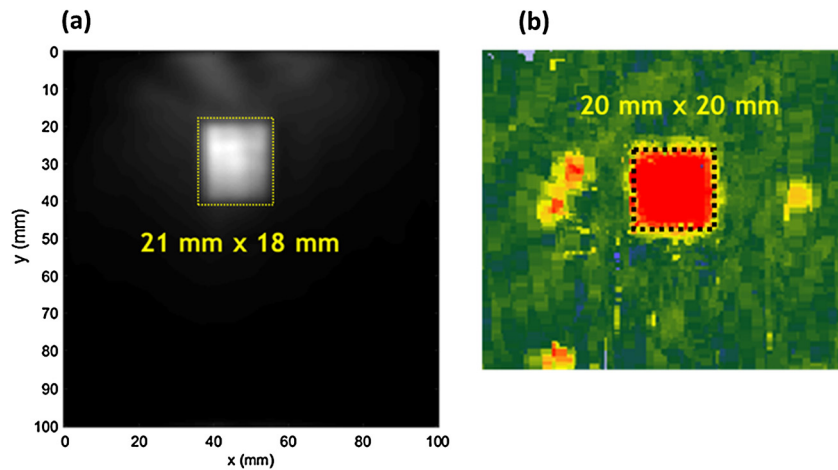


Fig. 9. Comparison between (a) the wavefield energy mapping using the proposed algorithm, and (b) C-scan result.

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